

Abstract Data Types

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In C we can hide data structures using opaque pointers. In Python, we achieve this using classes and encapsulation.

The stack (LIFO) is implemented through a list. While a queue (FIFO) is implemented through a deque.

```
from collections import deque
```

```
# STACK (Slides 65-77)
```

```
stack = []
```

```
stack.append(1) # Push (O(1))
```

```
stack.append(2)
```

```
top = stack.pop() # Pop (O(1)) -> Returns 2
```

```
# QUEUE (Slides 78-94)
```

```
queue = deque()
```

```
queue.append(1) # Enqueue (O(1))
```

```
queue.append(2)
```

```
first = queue.popleft() # Dequeue (O(1)) -> Returns 1
```